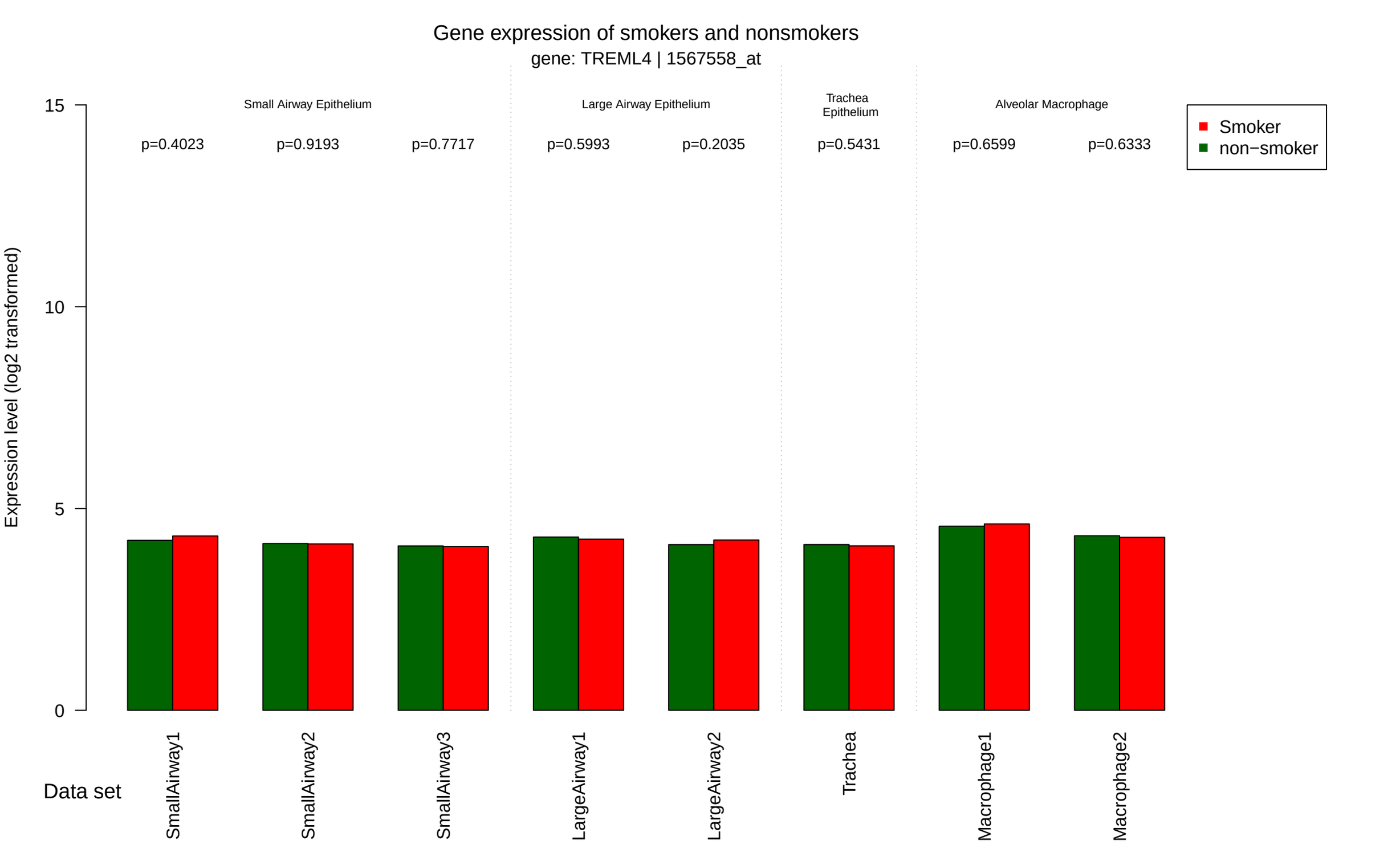
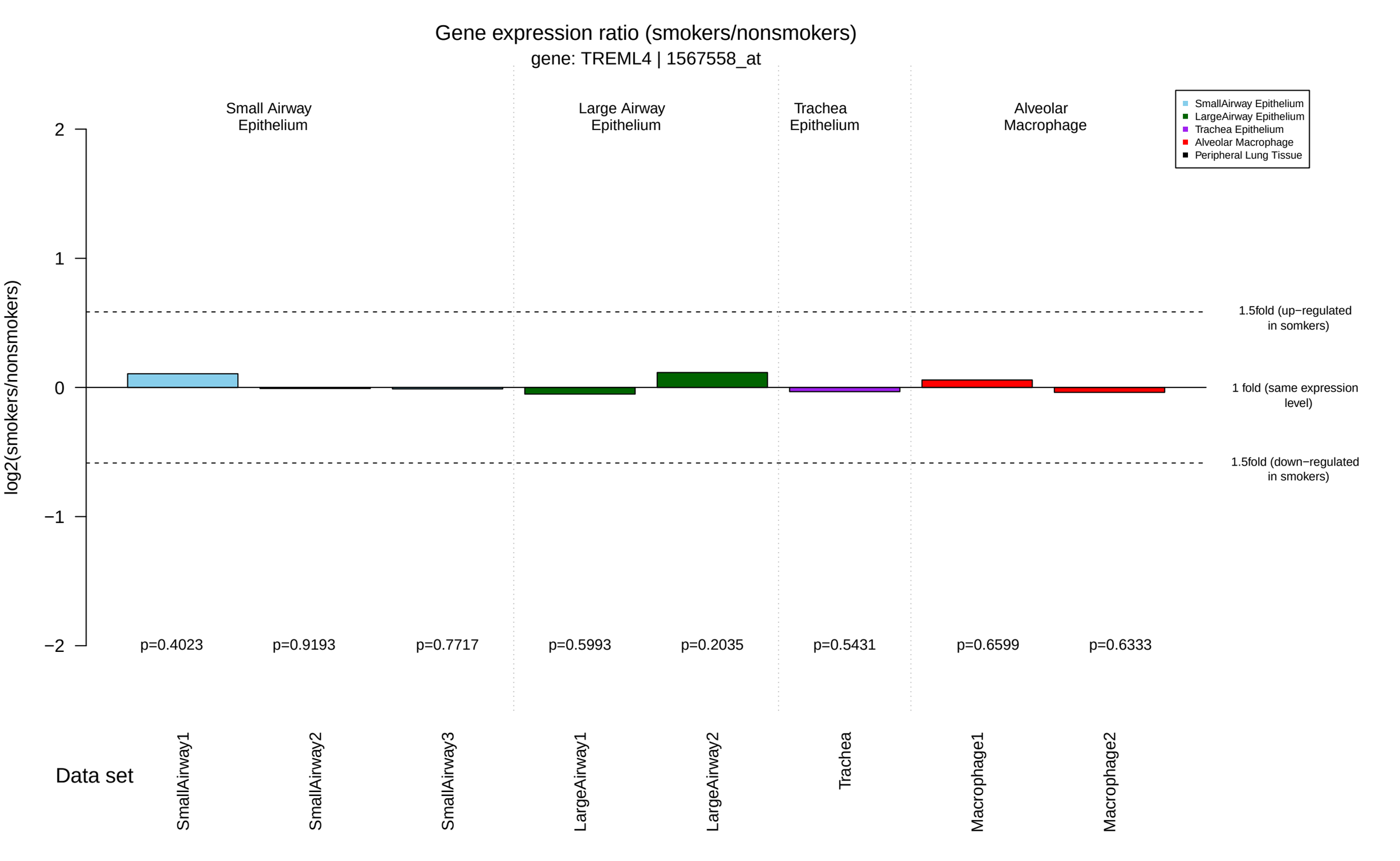
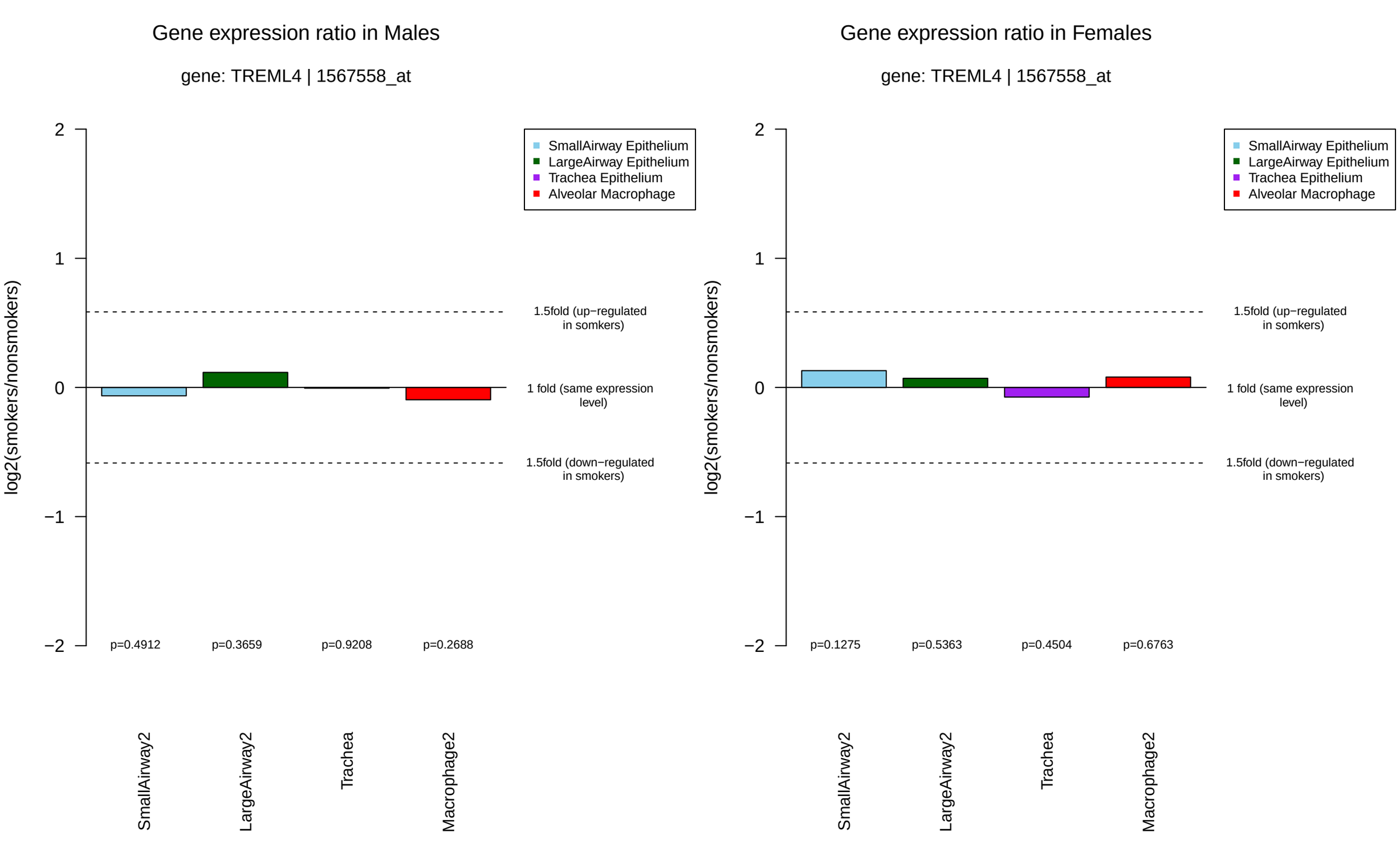
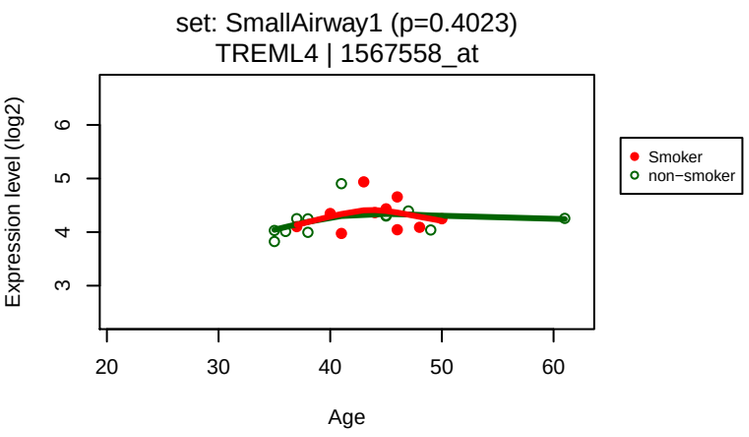
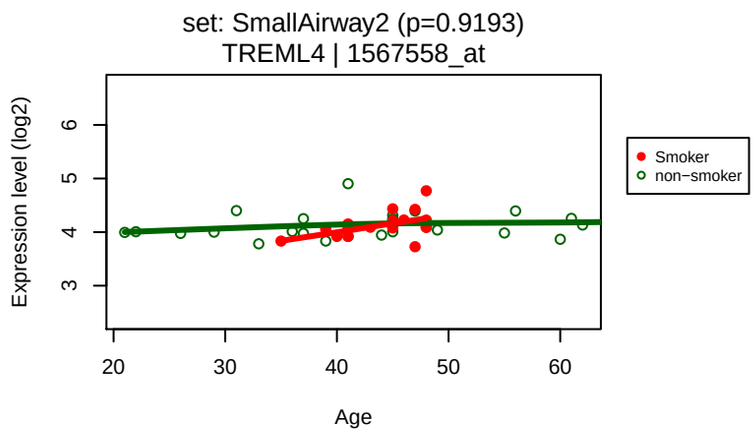
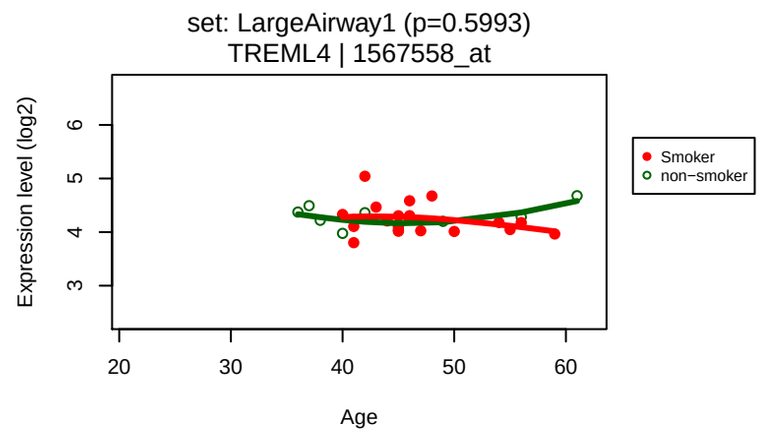

Gene expression of smokers and nonsmokers
gene: TREML4 | 1567558_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4023
p=0.9193
p=0.7717
p=0.5993
p=0.2035
p=0.5431
p=0.6599
p=0.6333
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

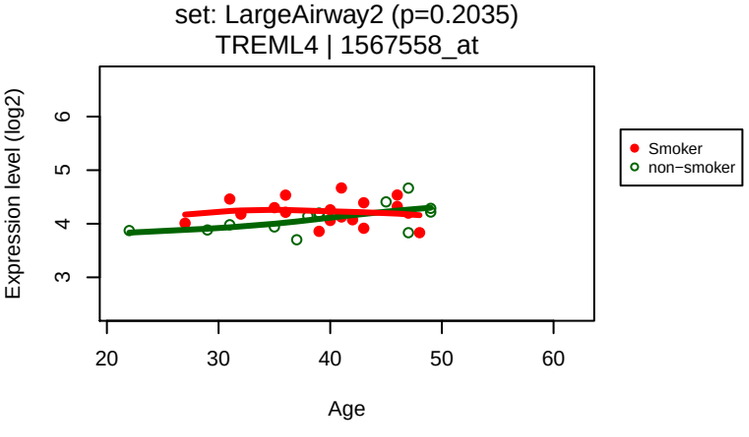

Gene expression ratio (smokers/nonsmokers)
gene: TREML4 | 1567558_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4023
p=0.9193
p=0.7717
p=0.5993
p=0.2035
p=0.5431
p=0.6599
p=0.6333
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

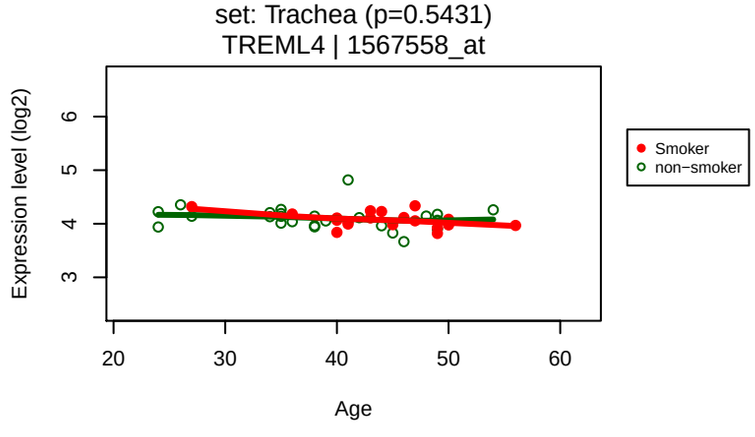

Gene expression ratio in Males
gene: TREML4 | 1567558_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4912
p=0.3659
p=0.9208
p=0.2688
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TREML4 | 1567558_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1275
p=0.5363
p=0.4504
p=0.6763
SmallAirway2
LargeAirway2
Trachea
Macrophage2

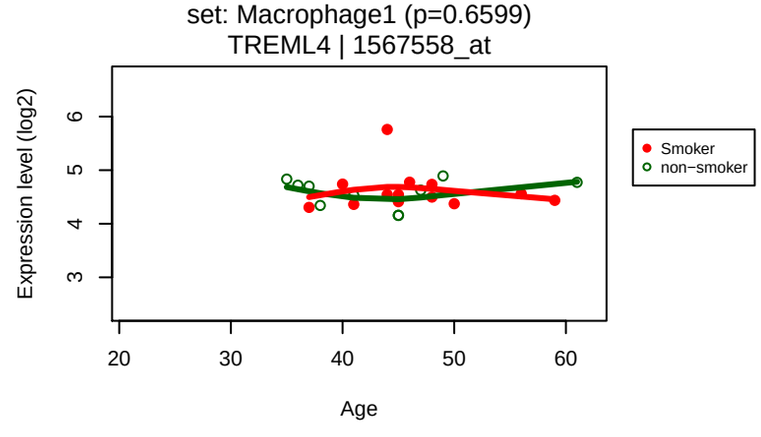

set: SmallAirway1 (p=0.4023)
TREML4 | 1567558_at
Expression level (log2)
Age
Smoker
non−smoker

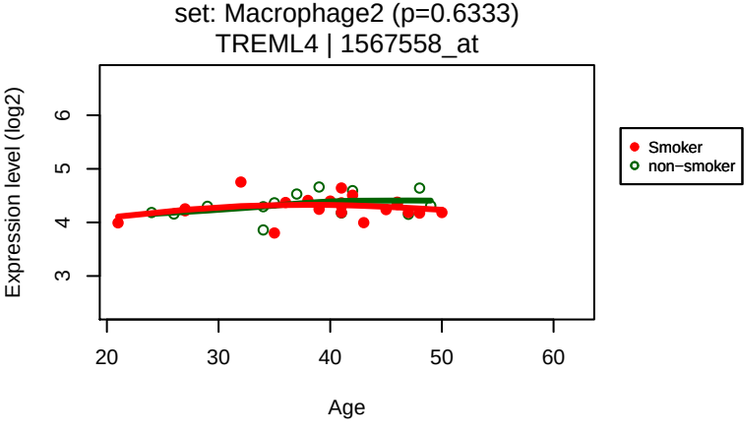

set: SmallAirway2 (p=0.9193)
TREML4 | 1567558_at
Expression level (log2)
Age
Smoker
non−smoker

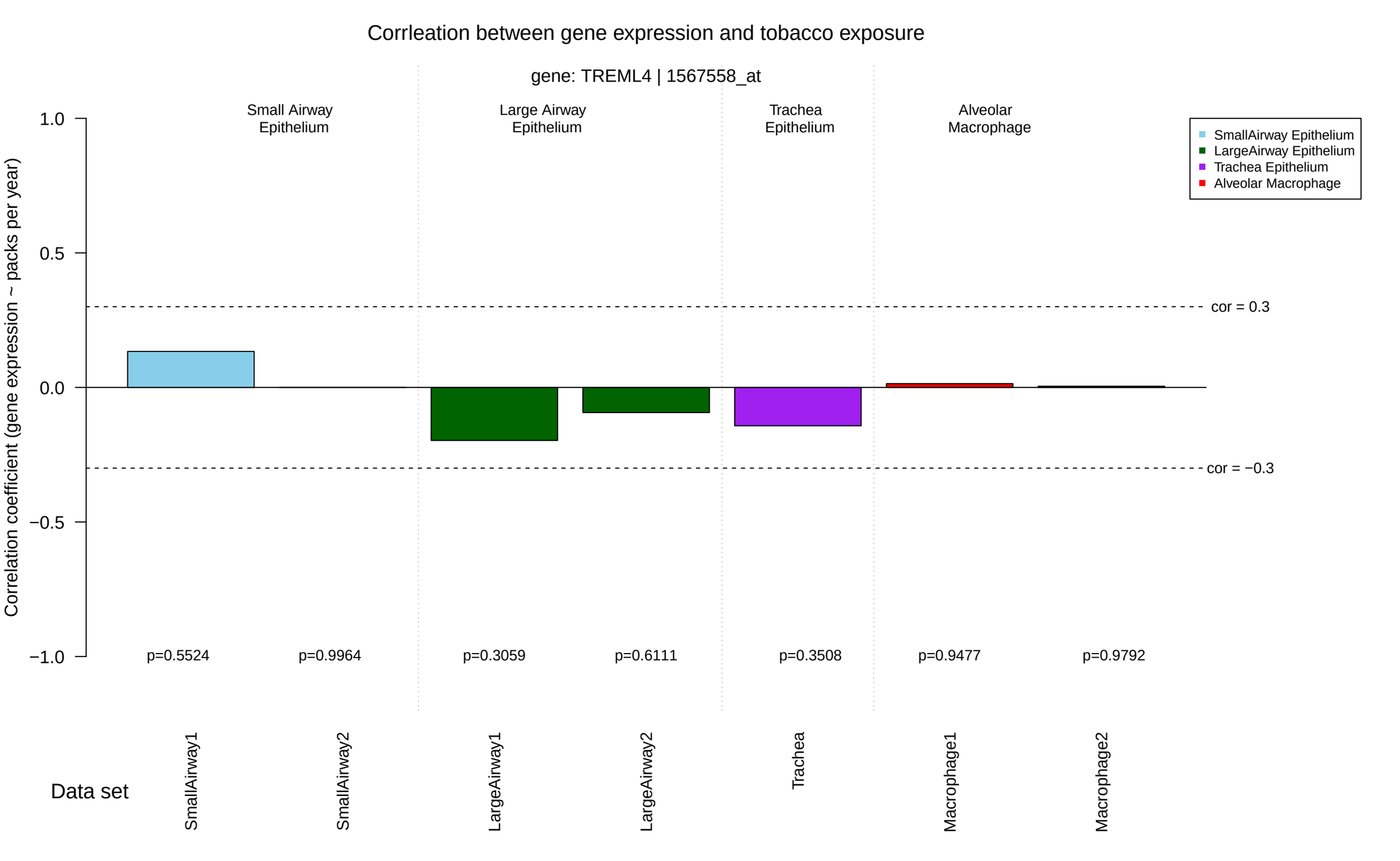

set: LargeAirway1 (p=0.5993)
TREML4 | 1567558_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2035)
TREML4 | 1567558_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5431)
TREML4 | 1567558_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6599)
TREML4 | 1567558_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6333)
TREML4 | 1567558_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TREML4 | 1567558_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5524
p=0.9964
p=0.3059
p=0.6111
p=0.3508
p=0.9477
p=0.9792
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2